

THE WAY IS OPEN



Segurança vs. performance no MySQL



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Support Engineer

Jean da Silva

- **Lead Database Performance Engineer**

O'Reilly – MongoDB Subject Matter Expert.

- MongoDB Fundamental – Interactive Labs.
- MongoDB Essentials – Video course.

Speaker :

- **MySQL Meetup Brazil** – Percona Toolkit na vida do DBA
- **Percona Webinar** – Improving MongoDB Index Performance
- **Percona Live ONLINE** – MongoDB Security Features.
- **Percona Live**– PBM – The Backup Open Source Tool for MongoDB.

Blog posts:

- [Benchmarking MongoDB do 3.6 ao 8.0-Parte 1](#)
- [Benchmarking MongoDB do 3.6 ao 8.0-Parte 2](#)
- [MongoDB Log and the Message “RSM Not Processing Response”](#)
- [Configuring Percona Backup for MongoDB in a Multi-Instances Environment](#)

[...]





Agenda

01 Motivação e Objetivo

03 Metodologia de teste

05 Conclusão

02 Ambiente de teste

04 Resultados



Motivação



Security Vulnerabilities

Database privilege abuse

Examples:

- Violation of least-privilege
- (PoLP)
- Stolen credentials

Network snooping

Examples:

- Packet sniffing
- IP/DNS spoofing

Data theft

Examples:

- Lost or stolen physical storage media
- Database file exposed

Access memory of database host

Examples:

- Memory dump analysis
- RAM scraping

"Encrypting MySQL Database Connections (Data in Motion)"

- **TLS 1.3** optimizes connection negotiation, **reducing overhead** and improving speed compared to TLS 1.2. Its optimized cipher suites provide stronger cryptography with **slightly better performance**

"Encrypt Data-at-rest Using Transparent Data Encryption"

- Another concern when **encrypting** data is the **performance impact** on database and application operations.

"Strategic Database Auditing"

- **Auditing** all database activity is impractical **due to storage costs, performance impacts,** and the risk of obscuring malicious events.



Quanto será que eu perco de performance por querer um banco mais seguro?... **Perde mesmo?**

— **Eu mesmo.**

Objetivo



Ordem de Grandeza; Algo tangível.

Obter métricas, comparar e avaliar.

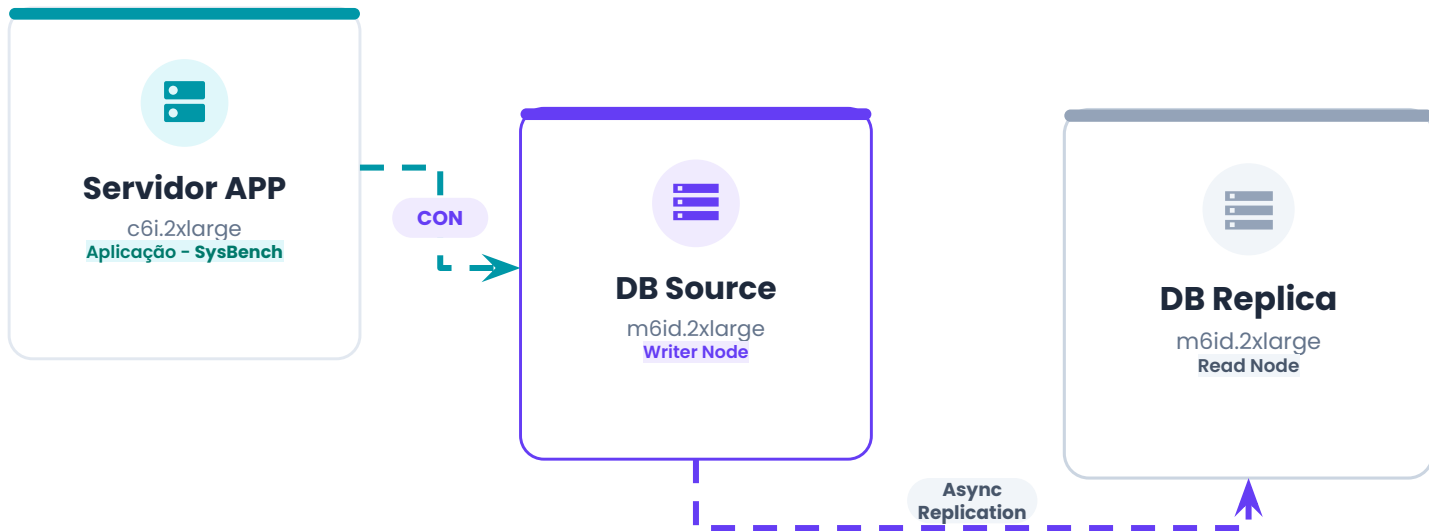
Avaliar o impacto com resultados práticos obtidos em testes.

Observação!

**Ambiente
de teste**



Topologia



Servidores

2x Servidor Banco de Datos

Sistema:	Amazon EC2; m6id.2xlarge
CPU:	physical = 1 cores = 4 virtual = 8 hyperthreading = yes
RAM:	32GB
Discos:	441.5 GB nvme2n1 XFS -> datadir 50 GB EBS XFS -> SO + logs
SO:	Red Hat 8.10 (Ootpa)
Kernel:	4.18.0-553.120.1.el8_10.x86_64

Servidor APP

Sistema:	Amazon EC2; c6i.2xlarge
CPU:	physical = 1 cores = 4 virtual = 8 hyperthreading = yes
RAM:	16GB
Discos:	50 GB EBS XFS -> (SO, sysbench, PMM, scripts)
SO:	Red Hat 8.10 (Ootpa)
Kernel:	4.18.0-553.120.1.el8_10.x86_64

Configuração Sistema

- **THP** = *never*
- **Vm.swappiness** = *1*
- **Swap** = *4 GB* (*/swapfile* no root)
- **jemalloc** no *mysqld* (*LD_PRELOAD*)
- **OOMScoreAdjust** = *-800*
- **I/O scheduler** = *none* (NVMe EC2)
- **CPU governor** = *N/A* (EC2)

Configuração base do Banco

- Percona Server: **8.4.11-11**
- `Innodb_buffer_pool_size` = **20G**
- `Innodb_redo_log_capacity` = **2G**
- `Innodb_flush_method` = **O_DIRECT**
- `Innodb_flush_log_at_trx_commit` = **1**
- `Sync_binlog` = **1**

Configuração SysBench

- **Versão do sysbench:** 1.0.20 (LuaJIT 2.1)
- **Workload:** `oltp_read_write` (*SELECT, UPDATE, DELETE e INSERT na mesma transação*)
- **Concorrência:** 32 threads
- **Dataset:** 10 tabelas(*sbtest1...sbtest10*) × 1 milhão de linhas cada:

```
CREATE TABLE sbtest1 (  
  id INT NOT NULL AUTO_INCREMENT,  
  k INT NOT NULL DEFAULT '0',  
  c CHAR(120) NOT NULL DEFAULT '',  
  pad CHAR(60) NOT NULL DEFAULT '',  
  PRIMARY KEY (id),  
  KEY k_1 (k)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

Configuração SysBench

- **Workload:** `oltp_read_write` (`SELECT`, `UPDATE`, `DELETE` e `INSERT` na mesma transação)

Qtd	Type	Shape
10	point SELECT	<code>SELECT c FROM sbtestN WHERE id=?</code>
1	range	<code>SELECT c ... WHERE id BETWEEN ? AND ? (range_size=100)</code>
1	sum range	<code>SELECT SUM(k) ... BETWEEN</code>
1	order range	<code>SELECT c ... BETWEEN ... ORDER BY c</code>
1	distinct range	<code>SELECT DISTINCT c ... BETWEEN ... ORDER BY c</code>
1	UPDATE índice	<code>UPDATE ... SET k=k+1 WHERE id=?</code>
1	UPDATE não-índice	<code>UPDATE ... SET c=? WHERE id=?</code>
1	DELETE+INSERT	mesmo id

Metodologia de teste



**Melhores
Práticas**

**TLS/SSL
intra-nodes**

Audit Log

01

02

03

04

05

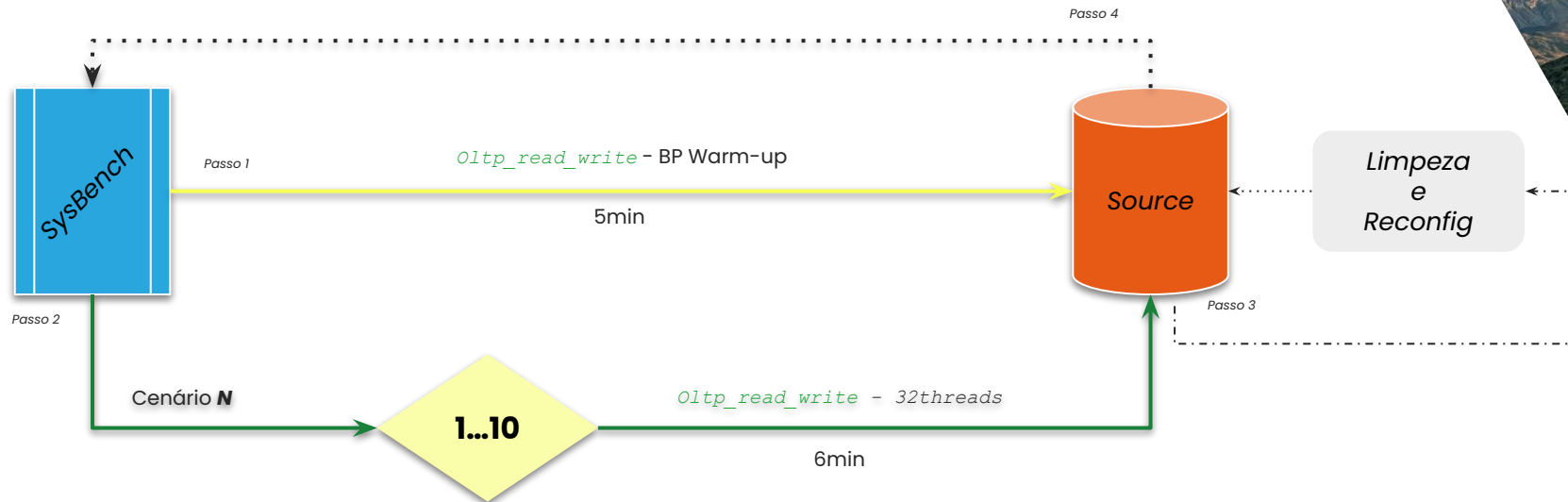
06

**TLS/SSL no
cliente**

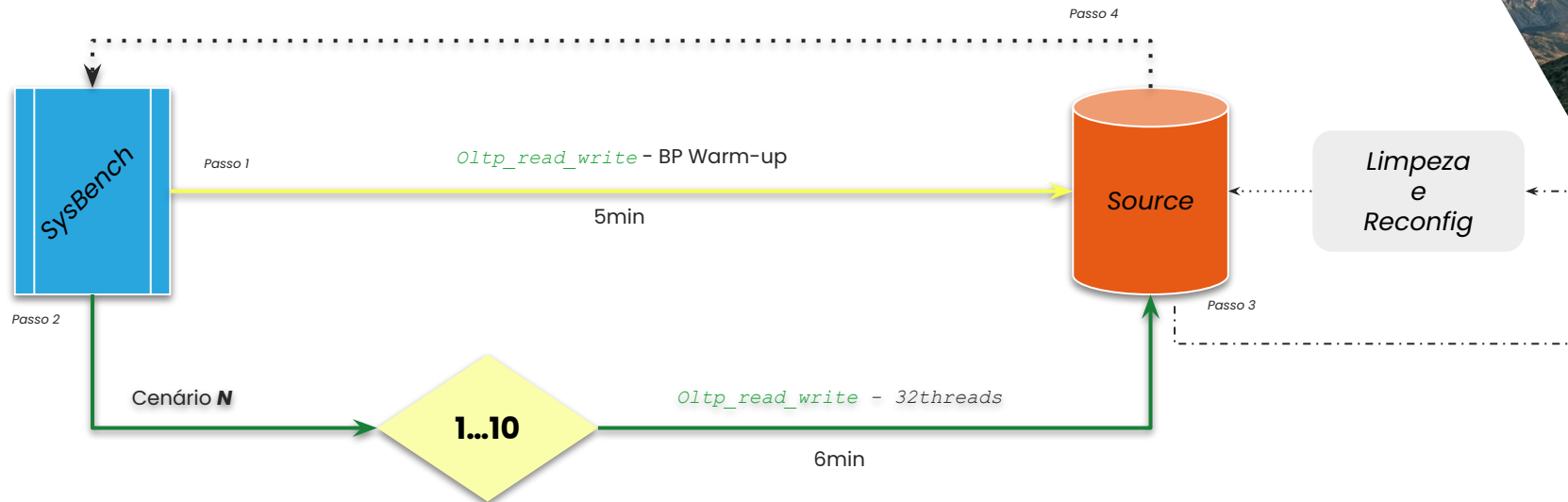
**Criptografia
em repouso**

Data Masking

Rotina



Rotina





Resultados



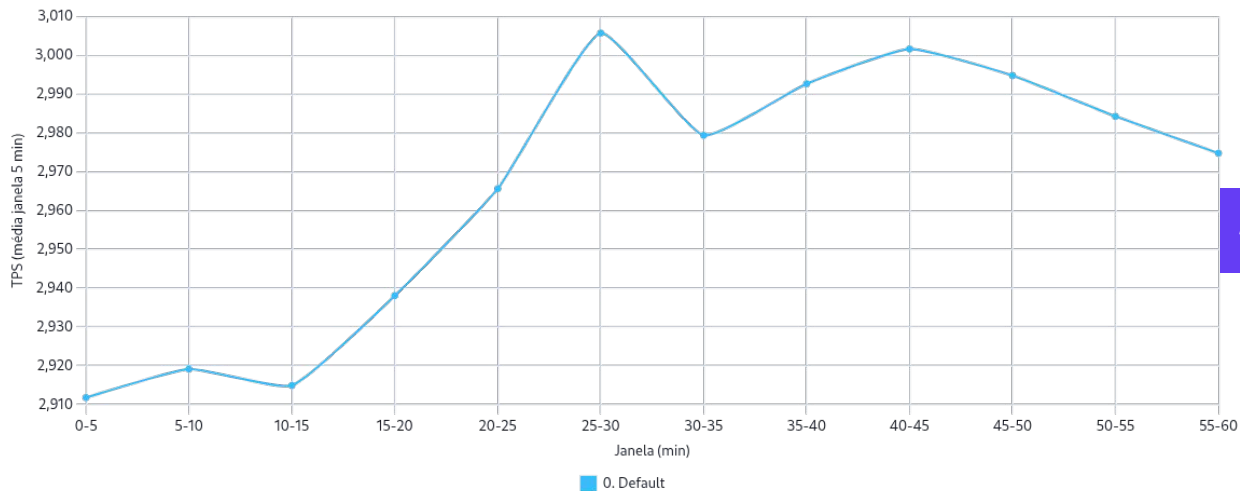
0. Deploy base

Banco utilizando o padrão
pós instalação

- BP 20G, redo 2G, `O_DIRECT`
- `innodb_flush_log_at_trx_commit = 1`
- `sync_binlog = 1`
- jemalloc + THP never + swappiness 1
- `bind-address = 0.0.0.0`
- `require_secure_transport = OFF`
- `local_infile = ON`
- `Secure_file_priv ' '`
- `default_password_lifetime = 0`
- Sem `validate_password`

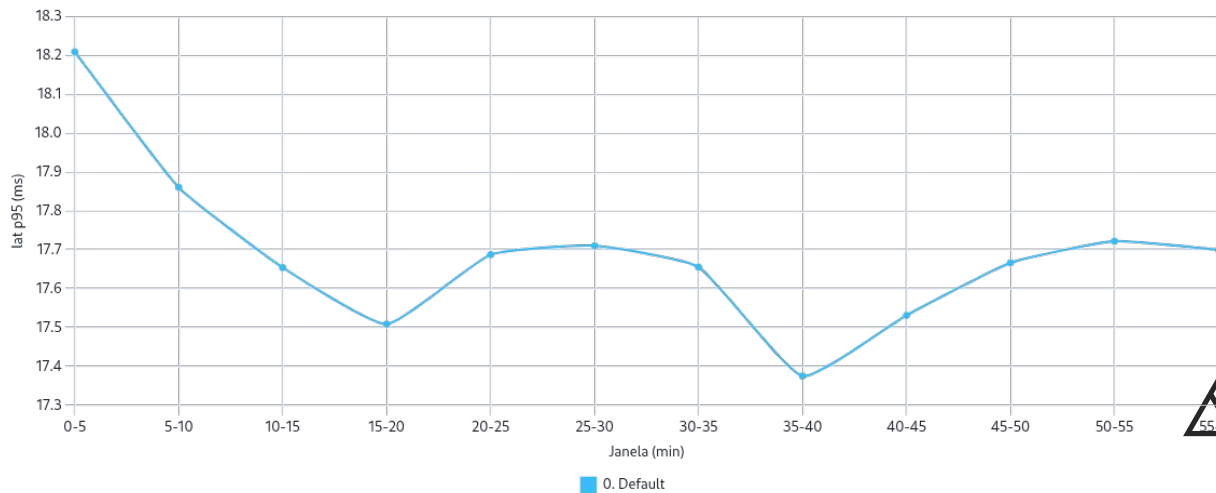
0. Deploy base - TPS

Estabilidade — TPS por janela de 5 min



~2966 TPS

Estabilidade — latência p95 por janela de 5 min

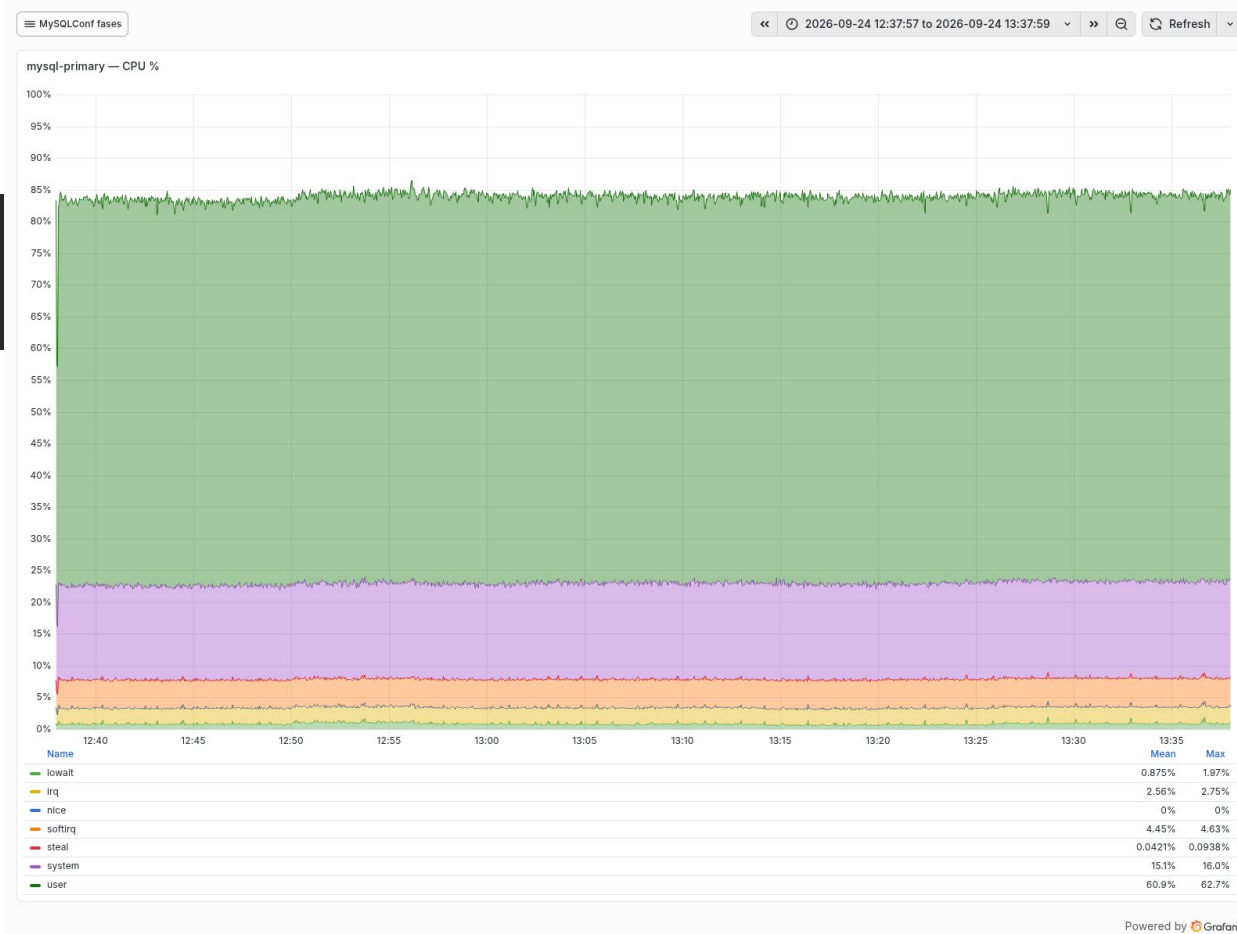


~17.17ms



PERCONA®

0. Deploy base - CPU



1. +Boas práticas de Segurança

Banco utilizando o padrão pós instalação + **boas práticas** de controle e segurança.

Inalterado

BP 20G, redo 2G, O_DIRECT

innodb_flush_log_at_trx_commit = 1

sync_binlog = 1

jemalloc + THP never + swappiness 1

+

Senha / contas

validate_password = MEDIUM (component)

default_password_lifetime = 90

REVOKE FILE

DROP anonymous users

DROP DATABASE test

+

Rede

bind-address = 127.0.0.1,<IP privado>

local_infile = OFF

secure_file_priv = /var/lib/mysql-files

skip_name_resolve = ON

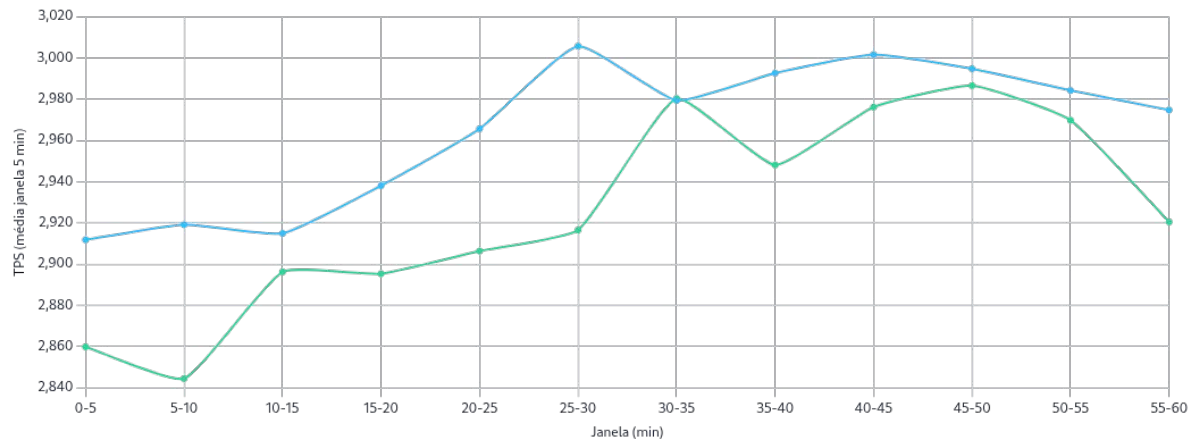
symbolic-links = OFF

automatic_sp_privileges = OFF

require_secure_transport = OFF

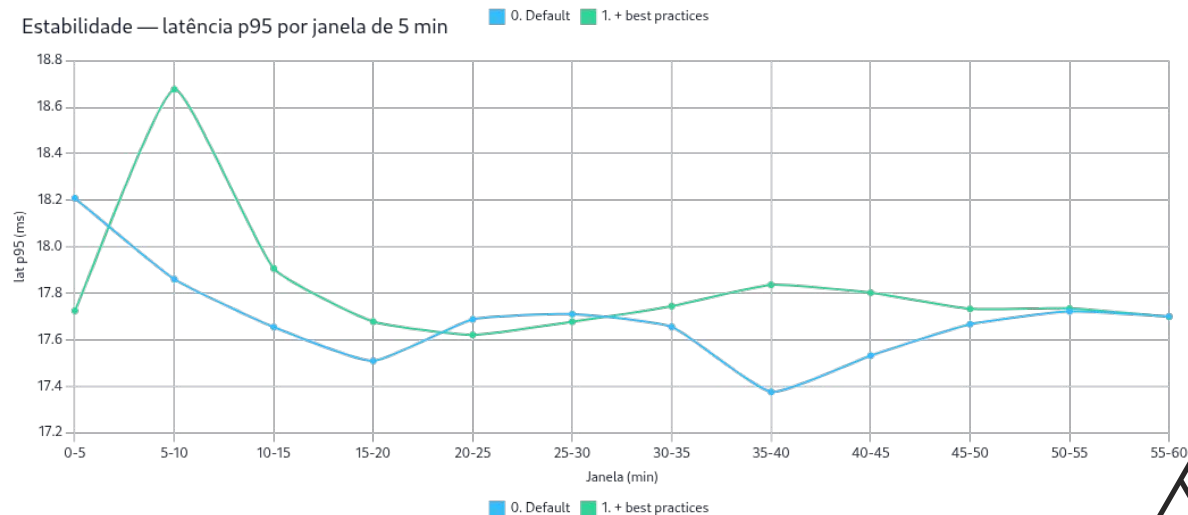
1. +Boas práticas de Sec - TPS

Estabilidade — TPS por janela de 5 min



~2925 TPS

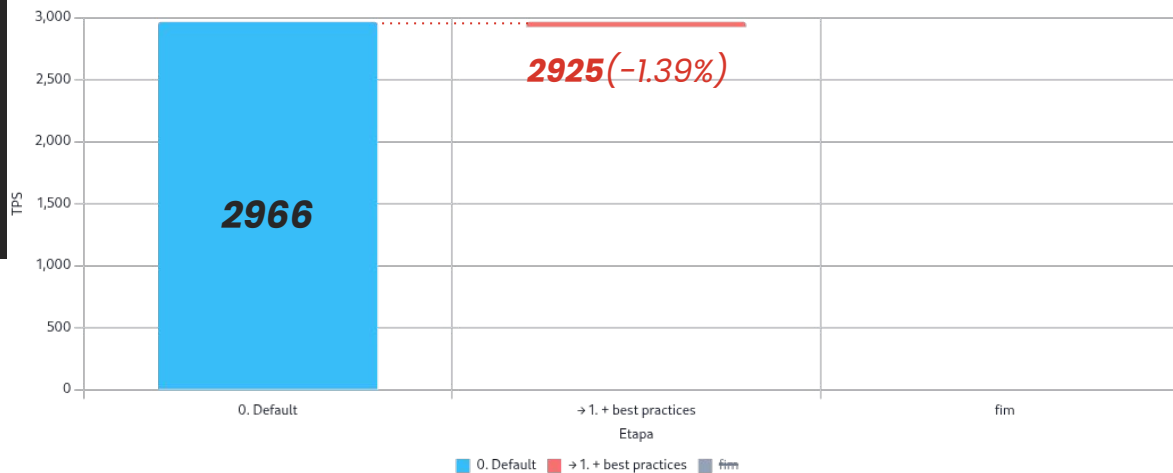
Estabilidade — latência p95 por janela de 5 min



~17.71ms

1. +Boas práticas de Sec - TPS

Waterfall — ganho / perda de TPS



2. +TLS/SSL no cliente

Adicionado
criptografia(TLS/SSL) entre
SysBench <-> Source node

```
## Inalterado ##  
BP 20G, redo 2G, O_DIRECT  
innodb_flush_log_at_trx_commit = 1  
sync_binlog = 1  
jemalloc + THP never + swappiness 1
```

+

```
## Senha / contas ##  
validate_password = MEDIUM (component)  
default_password_lifetime = 90  
REVOKE FILE  
DROP anonymous users  
DROP DATABASE test
```

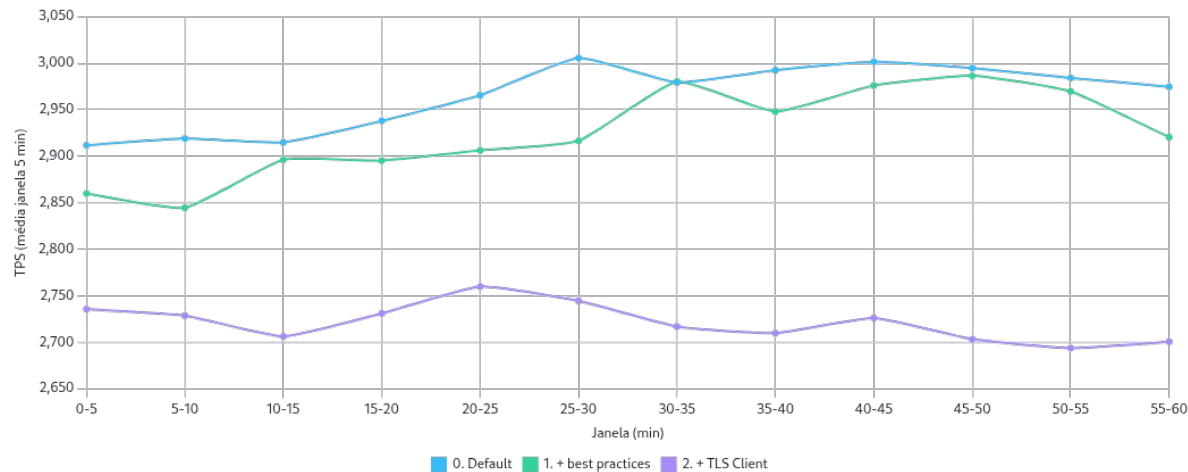
+

```
## Rede ##  
bind-address = 127.0.0.1,<IP privado>  
local_infile = OFF  
secure_file_priv = /var/lib/mysql-files  
skip_name_resolve = ON  
symbolic-links = OFF  
automatic_sp_privileges = OFF  
require_secure_transport = OFF
```

```
## TLS/SSL ##  
ALTER USER bench REQUIRE SSL  
sysbench com CA + client cert  
ssl_ca / ssl_cert / ssl_key nos dois  
nós  
require_secure_transport = OFF
```

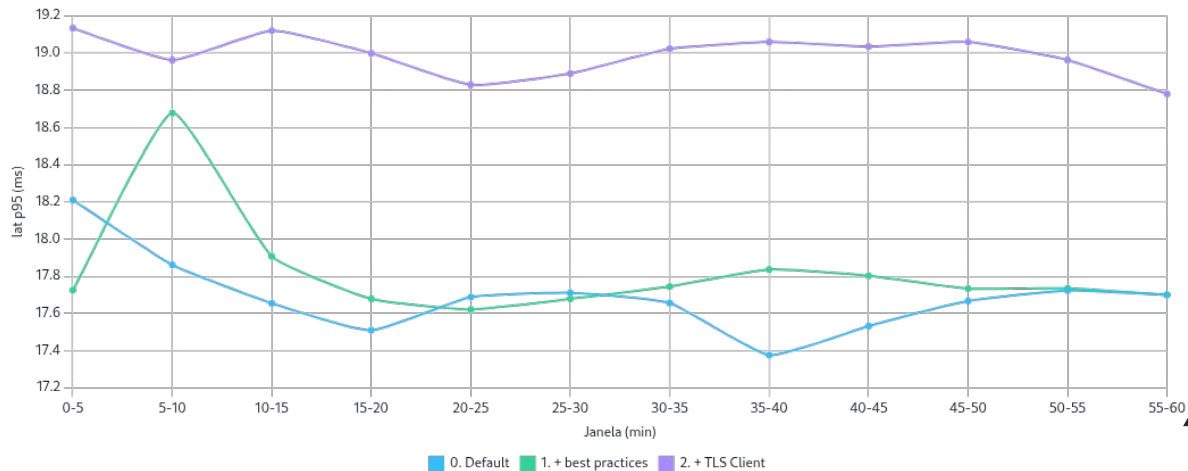
2. +TLS/SSL no cliente - TPS

Estabilidade — TPS por janela de 5 min



~2718 TPS

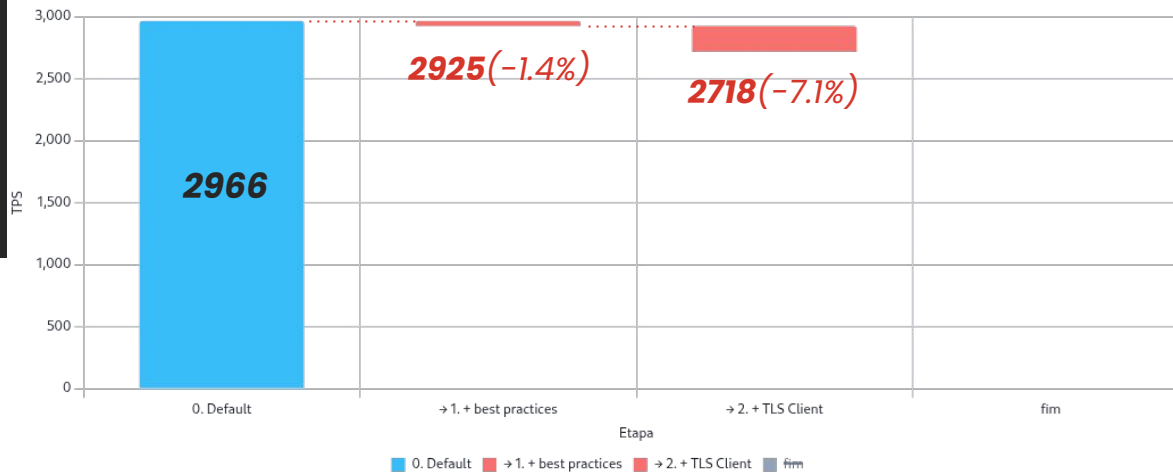
Estabilidade — latência p95 por janela de 5 min



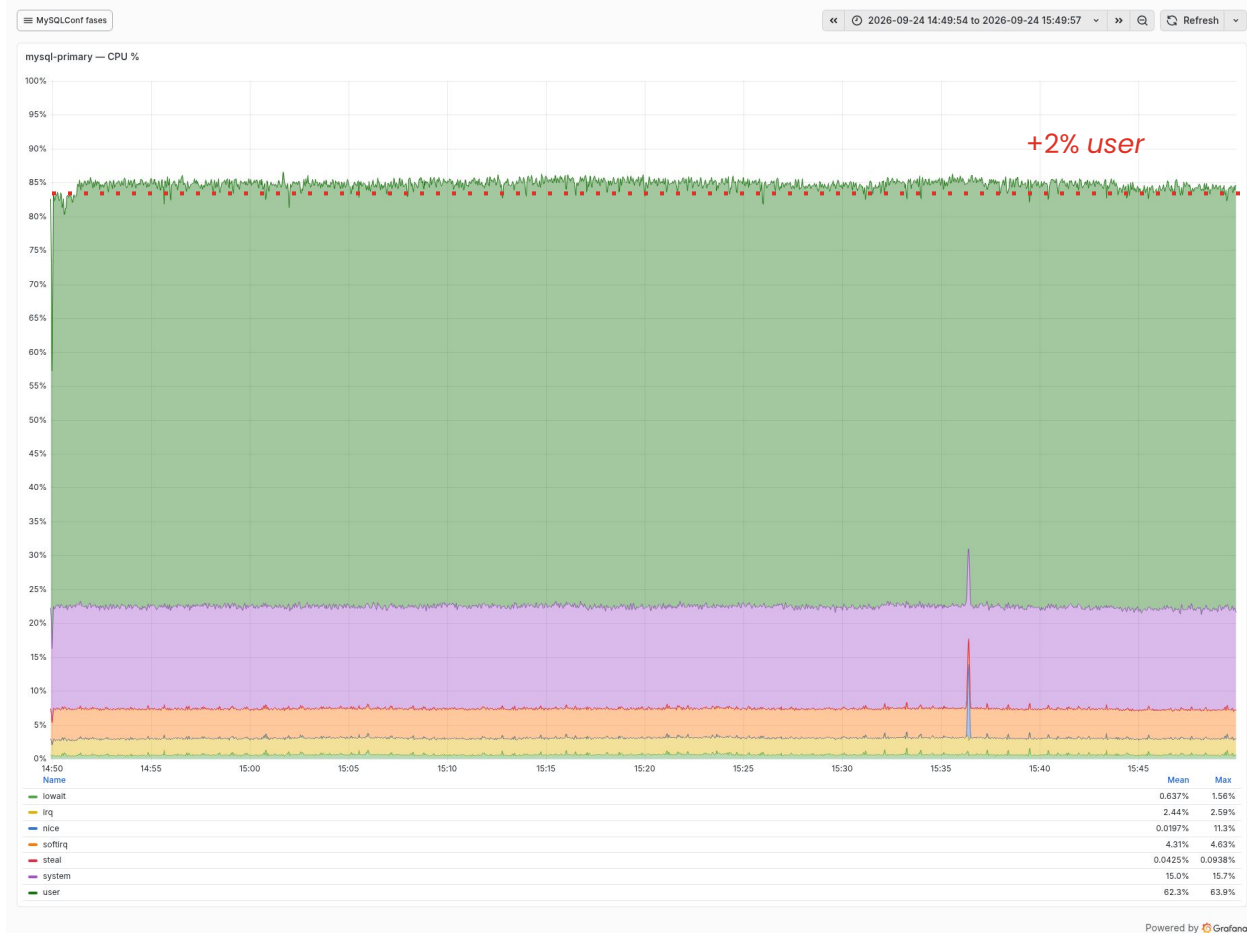
~19.04ms

2. +TLS/SSL no cliente - TPS

Waterfall — ganho / perda de TPS



2. +TLS/SSL no cliente - CPU



3. +TLS/SSL Intra nodes

Adicionado criptografia(TLS/SSL) entre Source <-> Replica + criptografia do binlog.

```
## Inalterado ##  
BP 20G, redo 2G, O_DIRECT  
innodb_flush_log_at_trx_commit = 1  
sync_binlog = 1  
jemalloc + THP never + swappiness 1
```

+

```
## Senha / contas ##  
validate_password = MEDIUM (component)  
default_password_lifetime = 90  
REVOKE FILE  
DROP anonymous users  
DROP DATABASE test
```

+

```
## Rede ##  
bind-address = 127.0.0.1,<IP privado>  
local_infile = OFF  
secure_file_priv = /var/lib/mysql-files  
skip_name_resolve = ON  
symbolic-links = OFF  
automatic_sp_privileges = OFF  
require_secure_transport = OFF
```

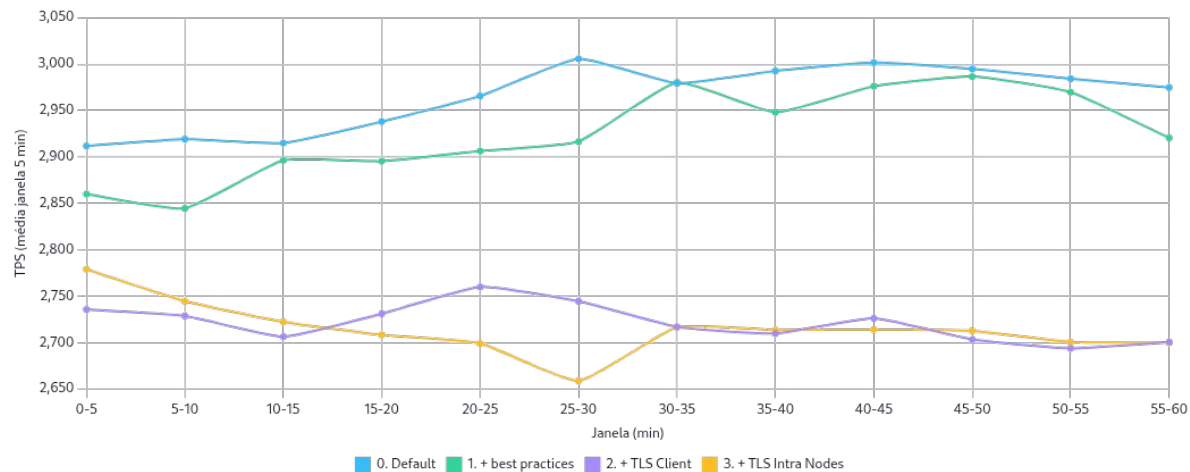
```
## TLS/SSL ##  
ALTER USER bench REQUIRE SSL  
sysbench com CA + client cert  
ssl_ca/ssl_cert/ssl_key nos dois nós
```

+

```
## TLS/SSL + binlog ##  
repl REQUIRE SSL  
SOURCE_SSL=1 + SOURCE_SSL_CA  
(réplica)  
require_secure_transport = ON  
binlog_encryption = ON (+ keyring)
```

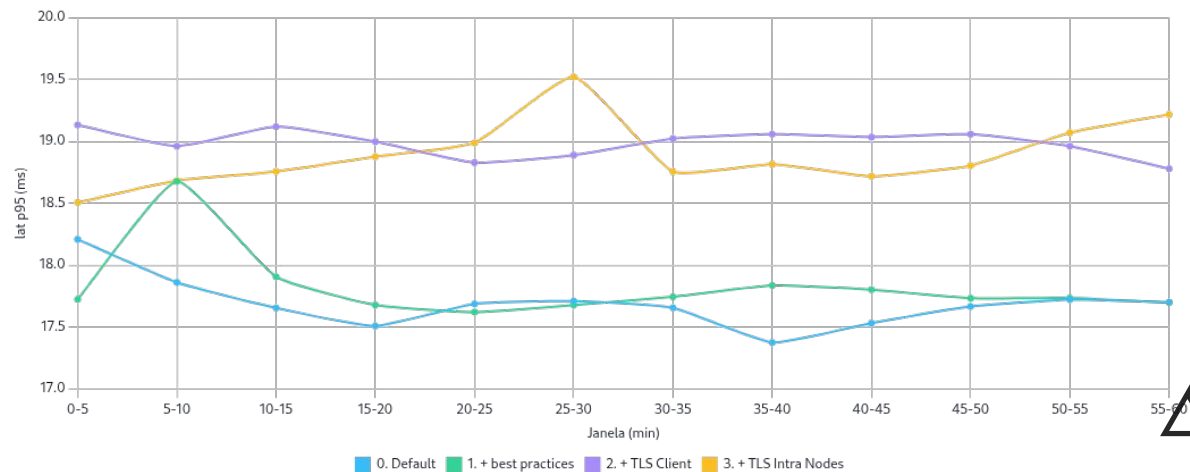
3. +TLS/SSL intra nodes - TPS

Estabilidade — TPS por janela de 5 min



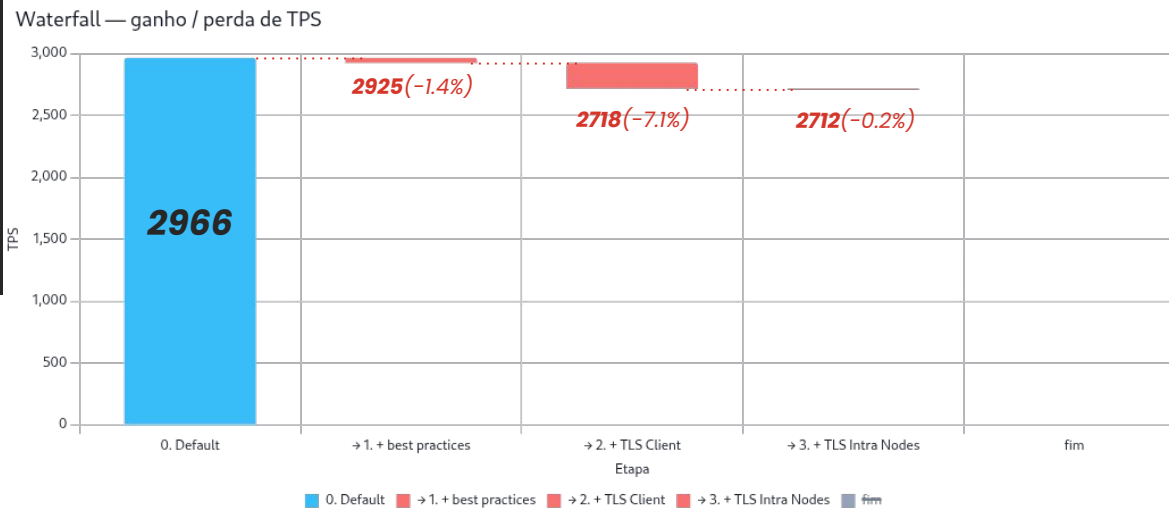
~2712 TPS

Estabilidade — latência p95 por janela de 5 min



~18.91ms

3. +TLS/SSL intra nodes - TPS



4. +TDE

Adicionado criptografia em disco.

```
## Inalterado ##  
BP 20G, redo 2G, O_DIRECT  
innodb_flush_log_at_trx_commit = 1  
sync_binlog = 1  
jemalloc + THP never + swappiness 1
```

+

```
## Senha / contas ##  
validate_password = MEDIUM (component)  
default_password_lifetime = 90  
REVOKE FILE  
DROP anonymous users  
DROP DATABASE test
```

+

```
## Rede ##  
bind-address = 127.0.0.1,<IP privado>  
local_infile = OFF  
secure_file_priv = /var/lib/mysql-files  
skip_name_resolve = ON  
symbolic-links = OFF  
automatic_sp_privileges = OFF  
require_secure_transport = OFF
```

```
## TLS/SSL ##  
ALTER USER bench REQUIRE SSL  
sysbench com CA + client cert  
ssl_ca/ssl_cert/ssl_key nos dois nós
```

+

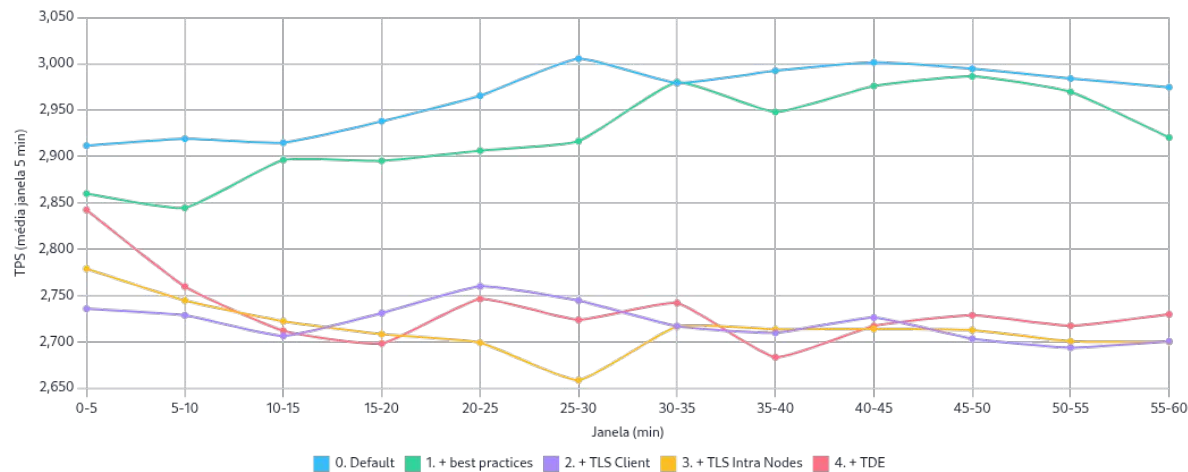
```
## TLS/SSL + binlog ##  
repl REQUIRE SSL  
SOURCE_SSL=1 + SOURCE_SSL_CA  
(réplica)  
require_secure_transport = ON  
binlog_encryption = ON (+ keyring)
```

+

```
## TDE ##  
innodb_redo_log_encrypt = ON  
innodb_undo_log_encrypt = ON  
innodb_temp_tablespace_encrypt = ON  
default_table_encryption = ON  
table_encryption_privilege_check = ON  
ALTER TABLE sbtest1..10  
ENCRYPTION='Y'
```

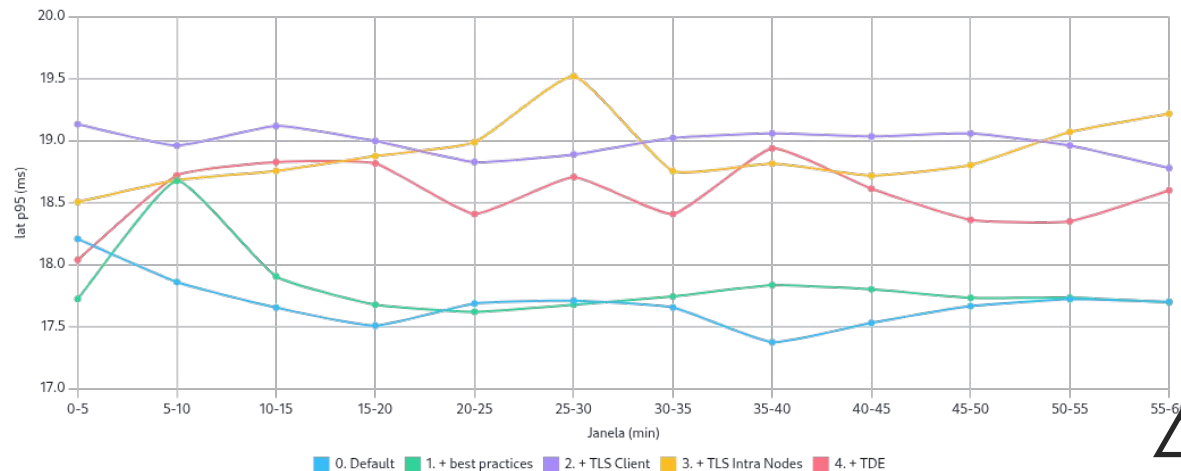
4. +TDE - TPS

Estabilidade — TPS por janela de 5 min



~2727 TPS

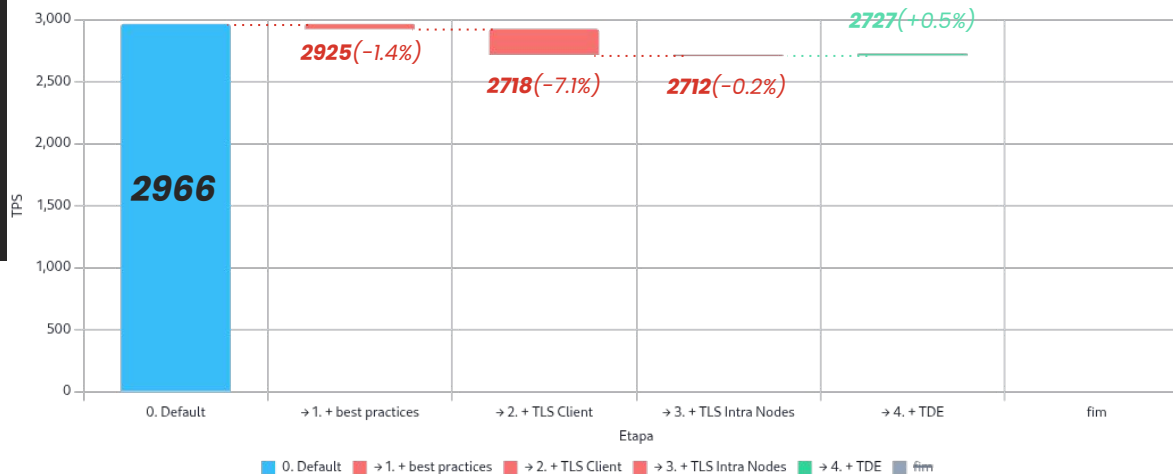
Estabilidade — latência p95 por janela de 5 min

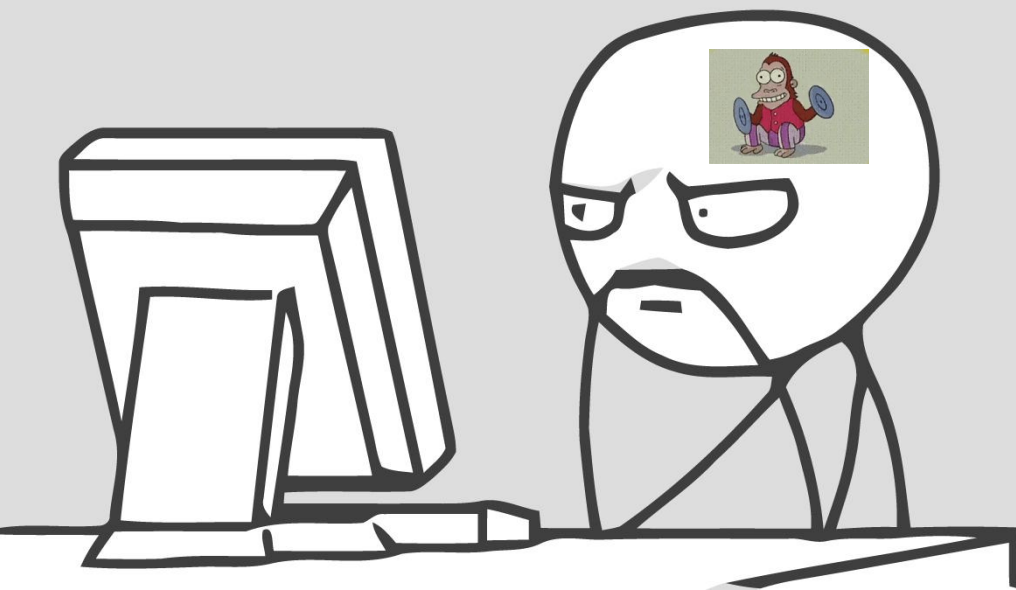


~18.53ms

4. +TDE - TPS

Waterfall — ganho / perda de TPS





5. +Audit log

Adicionado log de auditoria
em eventos específicos.

[...]

TLS/SSL

```
ALTER USER bench REQUIRE SSL  
sysbench com CA + client cert  
ssl_ca/ssl_cert/ssl_key nos dois nós
```

TLS/SSL + binlog

```
repl REQUIRE SSL  
SOURCE_SSL=1 + SOURCE_SSL_CA (réplica)  
require_secure_transport = ON  
binlog_encryption = ON (+ keyring)
```

Audit

```
component_audit_log_filter  
strategy = ASYNCHRONOUS  
rotate_on_size = 1G  
filtro log_security %  
· login recusado  
· DDL (create/alter/drop/truncate/rename)  
· GRANT / REVOKE
```

5. +Audit Log - TPS

Estabilidade — TPS por janela de 5 min



~2244 TPS

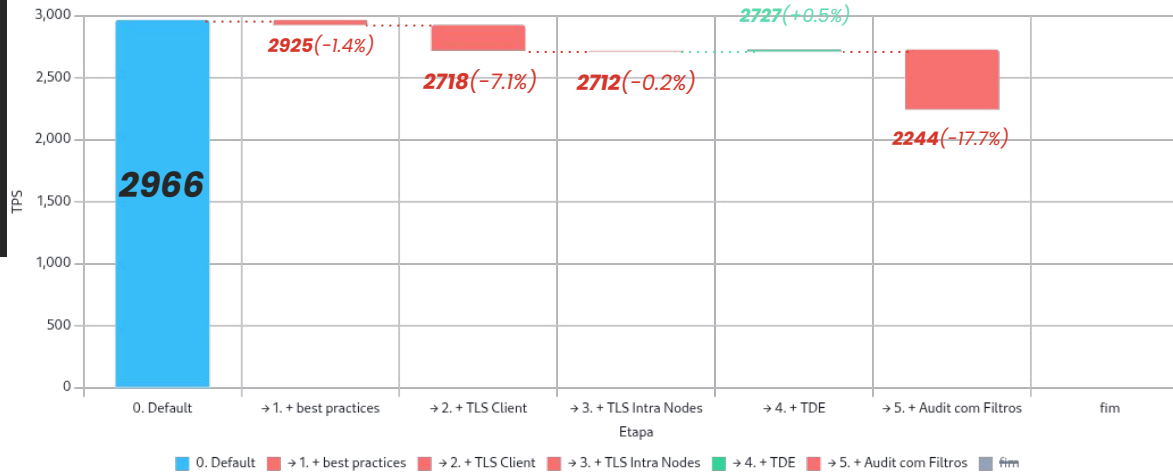
Estabilidade — latência p95 por janela de 5 min



~22.50ms

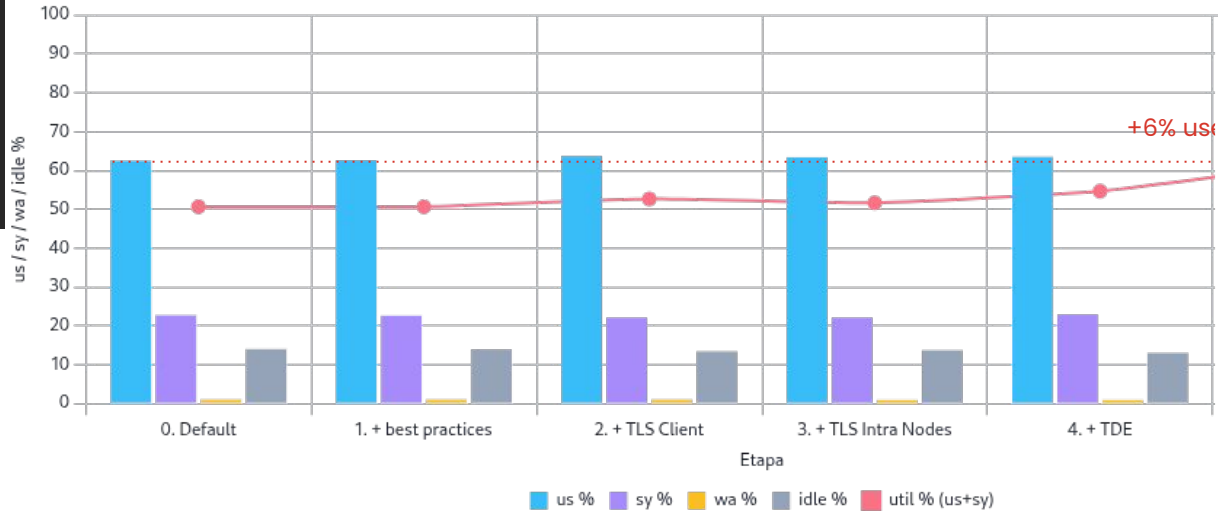
5. +Audit Log - TPS

Waterfall — ganho / perda de TPS



5. +Audit Log - CPU

CPU no primário (vmstat)



6. +Data Masking

Adicionado mascaramento de dados em **SELECTs**

```
[...]
```

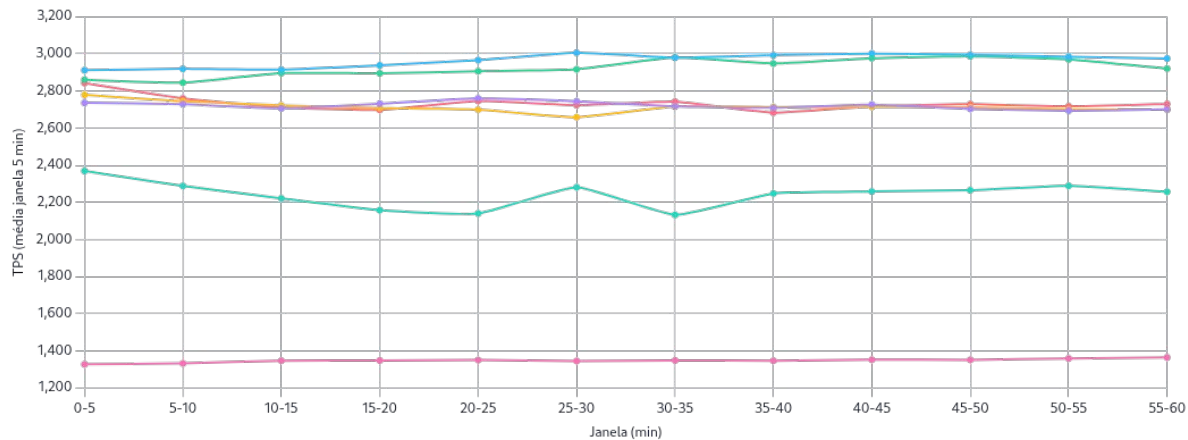
```
## TLS/SSL + binlog ##  
repl REQUIRE SSL  
SOURCE_SSL=1 + SOURCE_SSL_CA (réplica)  
require_secure_transport = ON  
binlog_encryption = ON (+ keyring)
```

```
## Audit ##  
component_audit_log_filter  
strategy = ASYNCHRONOUS  
rotate_on_size = 1G  
filtro_log_security %  
· login recusado  
· DDL (create/alter/drop/truncate/rename)  
· GRANT / REVOKE
```

```
## Data Masking ##  
component_masking_functions  
mysql.masking_dictionaries  
sysbench Option B: mask_inner(c, 2, 2) nos  
SELECTs
```

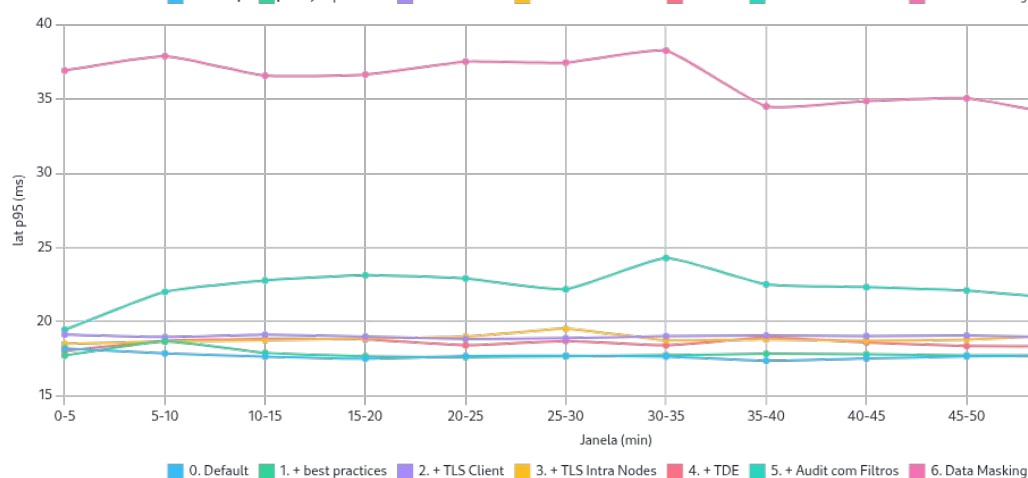
6. +Data Masking - TPS

Estabilidade — TPS por janela de 5 min



~2727 TPS

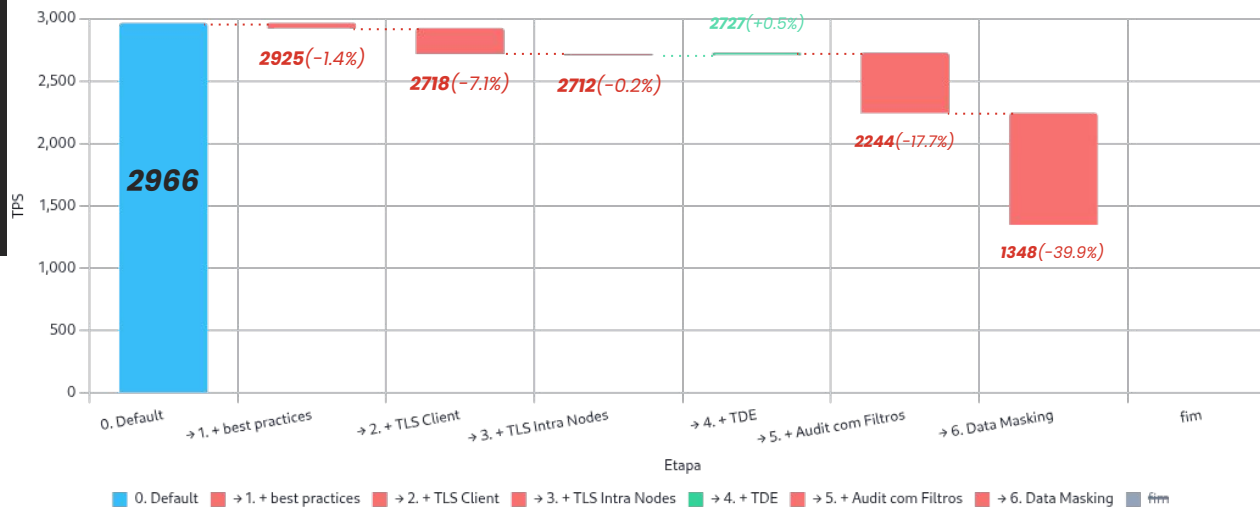
Estabilidade — latência p95 por janela de 5 min



~18.53ms

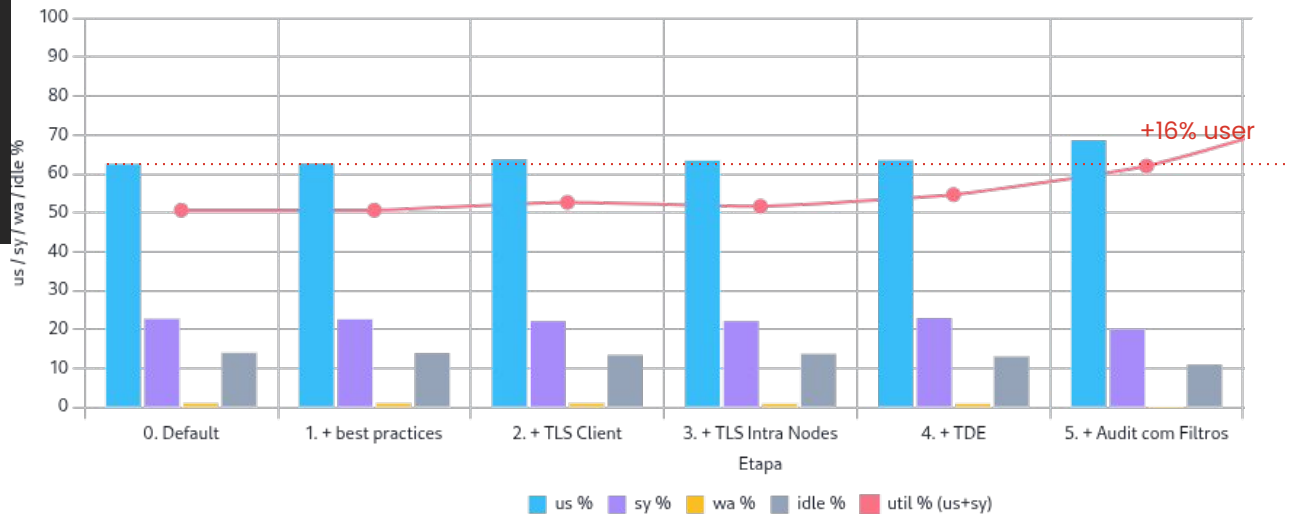
6. +Data Masking - TPS

Waterfall — ganho / perda de TPS



6. +Data Masking – CPU

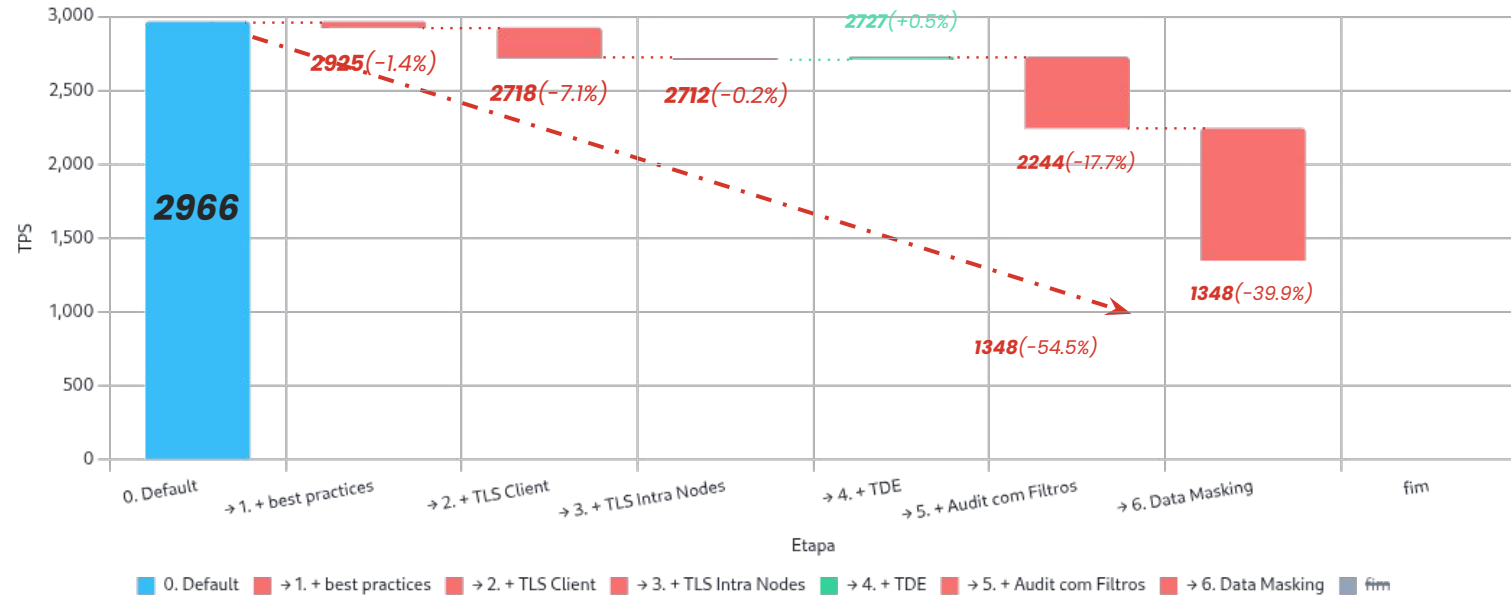
CPU no primário (vmstat)



**6. +Audit
Log – CPU**

Conclusão

Waterfall — ganho / perda de TPS



A full-page background image featuring a white goat with small horns standing on a sharp, snow-covered mountain ridge. The goat is looking towards the right. The background consists of a vast, rugged mountain range with numerous peaks and valleys covered in snow, under a clear, deep blue sky. The lighting suggests a low sun, casting long shadows and highlighting the textures of the snow and rock.

**THANK
YOU**

